Proof by Induction: Sum of Series

Questions 1–15

Question 1. Show that

a) $\frac{1}{3}k(k+1)(k+2) + (k+1)(k+2) = \frac{1}{3}(k+1)(k+2)(k+3)$

b) $\frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$

c) $(k+1)! - 1 + (k+1)(k+1)! = (k+2)! - 1$

Question 2. Prove by mathematical induction, for $n \in \mathbb{N}$:

a) $\sum_{r=1}^{n} 2^{r-1} = 2^n - 1$

b) $\sum_{r=1}^{n} 4^r = \frac{4}{3}(4^n - 1)$

Question 3. Prove by mathematical induction, for $n \in \mathbb{N}$:

a) $\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$

b) $\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$

c) $\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$

Question 4. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} r(r-1) = \frac{1}{3}n(n-1)(n+1)$.

Question 5. Prove by mathematical induction, for $n \in \mathbb{N}$, $-3-4-3+\cdots+n(n-4)=\frac{1}{6}n(n+1)(2n-11)$.

Question 6. Michael is partway through a proof, he writes: $\sum_{r=1}^{k+1} r(r+3)=\sum_{r=1}^{k} r(r+3)+(k+1)$. Identify his mistake and correct this line of working.

Question 7. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} \frac{1}{(r+1)(r+2)} = \frac{n}{2n+4}$.

Question 8. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} \frac{1}{(r+2)(r+4)} = \frac{n(7n+25)}{24(n+3)(n+4)}$.

Question 9. Prove by mathematical induction, for $n \in \mathbb{N}$:

a) $\displaystyle\sum_{r=1}^{n} \frac{1}{3^r} = \frac{1}{2} - \frac{1}{2 \times 3^n}$

b) $\displaystyle\sum_{r=1}^{n} \frac{r}{2^r} = 2 - \frac{n+2}{2^n}$

c) $\displaystyle\sum_{r=1}^{n} (r+1)2^r = n \times 2^{n+1}$

Question 10. It is conjectured that for constants a, b and c, $S_n = \displaystyle\sum_{r=1}^{n} (r+1)^2 = \frac{1}{6}n(an^2 + bn + c)$.

a) By considering S_1, S_2 and S_3 show that $a = 2$, $b = 9$ and $c = 13$.

b) Prove the conjecture by induction.

Question 11. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n}(r+i) = -n + \frac{1}{2}n(n+1)i$.

Question 12. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} r \times r! = (n+1)! - 1$.

Question 13. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} \frac{r}{(r+1)!} = 1 - \frac{1}{(n+1)!}$.

Question 14. Prove, for $n \in \mathbb{N}$ and $a \in \mathbb{R}$, $\sum_{r=1}^{n} \frac{1}{(r+a)(r+a+1)} = \frac{n}{(1+a)(n+1+a)}$.

Question 15. Prove, for $n \in \mathbb{N}$ and $a \in \mathbb{R}$, $\sum_{r=1}^{n} a^r = \frac{a(a^n - 1)}{a - 1}$.